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END SEMESTER EXAMINATION JUNE- 2025

Name of The Program: **B.Tech (Petroleum Engineering)**
Name of The Course: **Petroleum Production Engineering-II**

Semester: VI
Course Code: **TPE-605**

Time: 3 Hours

Maximum Marks: 100

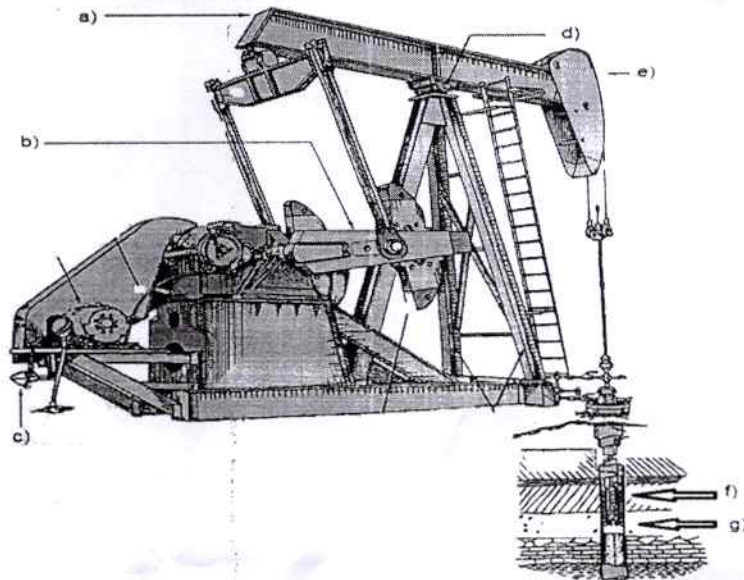
Note:

- (i) All Questions are compulsory
- (ii) Answer any two questions among a, b & c in each main question.
- (iii) Total marks each main question is twenty.
- (iv) Each Question carries 10 Marks

Q1.

(20 Marks)
(CO1,4)

- a) Explain the load which are temporary in nature/on platform.
- b) Describe the platform which rests on 4-6 pillar-like legs.
- c) From the given figure identify & explain type of Artificial lift and also identify parts (a, b, c, d, e, f, g) in the given figure, also write why it is used as artificial lift.



Q2.

(20 Marks)
(CO2,4,5)

- a) Explain the concept of offshore process equipment and open hole completion.
- b) Define EOR and also describe thermal method of EOR.
- c) Write 5-5 Advantages and disadvantages of Gas lift, Rod Pumping and ESP

Q3.

(20 Marks)
(CO3)

- a) Construct IPR of a vertical well in a saturated oil reservoir using Vogel's equation. The following data are given:

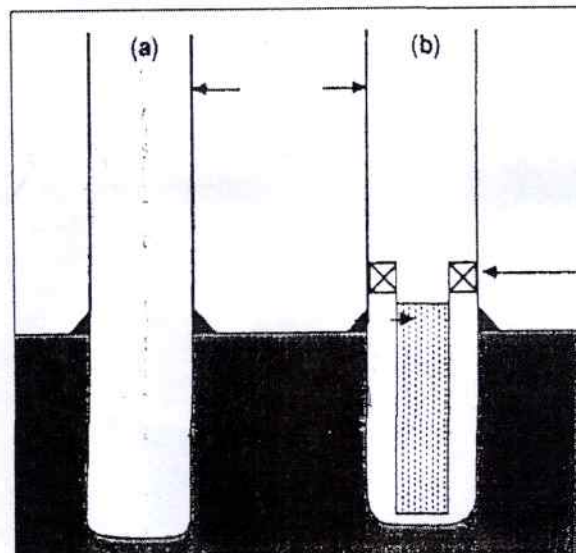
Effective horizontal permeability=8.2 md
 Pay zone thickness=53 ft
 Reservoir pressure=5651 psia
 Bubble-point pressure=5561 psia
 Fluid formation volume factor=1.1
 Fluid viscosity=5 1.7 cp
 Total compressibility=0.000129 psi⁻¹
 Drainage area=640 acres ($r_e=2980$ ft)
 Wellbore radius $r_w=0.328$ ft
 Skin factor=0

- b) Suppose that 1000 bbl/day of 40°API, 1.2 cp oil is being produced through $2\frac{7}{8}$ in., 8.6-lb_m/ft tubing in a well that is 15° from vertical. If the tubing wall relative roughness is 0.001, calculate the pressure drop over 1000 ft of tubing.
- c) Illustrates the nodes in an oil and gas production system and pressure trend along the flow path and also write all the abbreviations used for making nodes. and also describe what is nodal analysis and it is used in Petroleum Production.

Q4.

(20 Marks)
(CO2,5)

- a) From the given figure identify types of liner and differentiate between them & also identify & name the parts of liner which are shown by arrows.



- b) Describe the concept of oil processing module and drill ships.
- c) Differentiate Between Gas injection and Water Injection

Q5

(20 Marks)
(CO2,3,4)

- a) Explain the concept of abrasive perforation.
- b) Differentiate between wellbore performance & choke performance.
- c) Explain the artificial lift method which can be used for volumes range from a low of 150 B/D.